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Evaluating Architectural Trade-offs in Retrieval-Augmented Generation Systems for Financial Question Answering

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ABSTRACT

Retrieval-augmented generation is increasingly being used for financial question aswering where answers must be gorunded in complex documents without excessive response time. This study was created to examine how retrieval architecture affects evidence retrieval, answer quality, faithfulness and latency in financial QA. There were four system architectures that were evaluated on FinanceBench dataset: a closed-book answering (P0), dense retrieval (P1), hybrid dense and BM25 retriveal (P2) and hybrid retrieval with cross-encoder rereanking(P3). five of the 150 FinanceBench questions were used for smoke test while the remaining 145 formed the evaluation yielding 580 saved answers. All architectures used the same answer model. Evaluation covered annotated-page retrieval, answer overlap and numeric entity measures, model juded evidence support and per question latency.

P3 ranked annotated evidence highest at rank one (Hit@1 = 0.214), but matched P1 at rank five (Hit@5 = 0.400). P2 achieved the highest numeric-entity accuracy (0.190), while P1 had the highest measured context support (0.850) this mean covered only 85 scored answers; the three RAG pipelines abstained on 40–43 percent of questions. P3 averaged 5.63 seconds per question(3.7 seconds more than P1), with no established context-support gain over P2. Dense retrieval is a provisional latency-conscious choice for this benchmark. Because support was judged by the answer model in separate calls and complete answers were not independently adjudicated, the results do not identify a definitive factual-correctness winner. Github: https://github.com/arinzefu/finrag-arch-eval.git

Keywords: Financial QA, artificial intelligence, natural language processing, retrieval-augmented generation, large language model

1. INTRODUCTION

Financial question answering is the process of answering questions about a company's financial position, performance, valuation, and future prospects using financial statements and related disclosures. Unlike general question answering, financial QA often requires more than locating a sentence in a document; it involves interpreting and analysing accounting categories, identifying the correct reporting period, comparing figures, and reasoning over ratios, profitability, growth, operating activities, and financing activities. Nissim and Penman **ref20** show that financial statement analysis is a structured process used in equity valuation, where current financial ratios and accounting information are analysed as indicators of future earnings, residual income, and equity payoffs. Therefore, financial QA is inherently demanding because a reliable answer must preserve numerical accuracy, accounting meaning, and the relationship between financial values.

Natural Language Processing (NLP) provides a potential solution to question-answering (QA) tasks by enabling computational systems to process, interpret, and generate human language. Early foundations for this field can be traced to Harris's work on distributional structure, which proposed that linguistic elements could be analysed according to the contexts in which they occur **ref21**. This principle later became influential in computational approaches to language representation. However, the ability of NLP-based systems to provide accurate and reliable answers must still be critically evaluated **ref1**. In recent years, NLP has advanced rapidly, particularly through the development of large language models (LLMs), which are trained on extensive datasets and can capture broad linguistic patterns and contextual relationships. The emergence of modern LLMs was significantly enabled by the Transformer architecture introduced by Vaswani et al. (2017), which replaced recurrent sequence processing with self-attention mechanisms, allowing models to capture long-range dependencies more effectively and enabling more efficient parallel training.

However, despite these advances, LLMs retain several important limitations. They can struggle with complex reasoning, their internal knowledge is constrained by the information available during training,

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and they may generate inaccurate or unsupported responses when prompts are ambiguous or when relevant information is absent from their training data. These limitations can reduce their effectiveness in providing reliable and precise answers, particularly in knowledge-intensive domains such as finance **ref17**.

One approach that has emerged to address these limitations is Retrieval-Augmented Generation (RAG). Introduced by Lewis et al. (2020), RAG combines the generative capabilities of pretrained language models with information retrieved from an external knowledge source **ref5**. Rather than relying solely on knowledge encoded during model training, a RAG system retrieves relevant document passages from a knowledge base and uses this evidence to support the generation of more accurate and contextually grounded responses.

This is particularly important because LLMs can struggle with domain-specific questions, up-to-date information, and factual accuracy, sometimes producing unsupported or hallucinated responses when relevant knowledge is absent from their training data. By grounding generated answers in retrieved external sources, RAG can improve factual reliability and make LLM-based systems more suitable for knowledge-intensive tasks **ref19**.

Gao et al. **ref19** describe three types of RAG systems: naive RAG, advanced RAG, and modular RAG. Naive RAG represents the earliest form of RAG system and gained prominence after the widespread adoption of ChatGPT. Advanced RAG focuses on correcting earlier limitations of naive RAG, while modular RAG offers a higher level of adaptability and versatility than the previous two approaches. Retrieval-augmented generation is often used to reduce hallucinations by grounding LLM responses in external documents. However, the quality of a RAG system is strongly shaped by its retrieval architecture **ref10**.

One approach to evaluating the retrieval component of RAG systems is the Benchmarking Information Retrieval (BEIR) framework **ref10**. BEIR demonstrates that retrieval methods do not generalise equally well across domains and that no single retrieval approach consistently outperforms others across all datasets. Consequently, dense, sparse, hybrid, reranking, and table-aware retrieval methods may involve different trade-offs in retrieval effectiveness, generalisability, computational cost, and response latency.

Existing financial RAG studies often propose or optimise a single pipeline. However, fewer studies isolate the contribution of retrieval architecture under controlled conditions while also considering latency as an explicit evaluation criterion. This is important because a highly accurate architecture may still be impractical for financial question answering if its response latency is too high.

This paper compares four financial question-answering configurations: a closed-book baseline and three retrieval-augmented pipelines. It is a design-selection study of complete configurations, rather than a component-level ablation. Although the answer model, question set, corpus, and final context size were controlled, the retrieval methods and candidate depths differed between configurations. The findings therefore describe the performance of these implemented systems and cannot isolate the causal effect of any single retrieval component. The contributions are:

- A comparison of four financial QA configurations on a common held-out FinanceBench question set.
- A joint evaluation of annotated-page retrieval, automated answer-overlap and numeric-entity measures, model-judged evidence support, abstention, and latency.
- A provisional architecture recommendation based on the observed quality–latency trade-off and its measurement limitations.

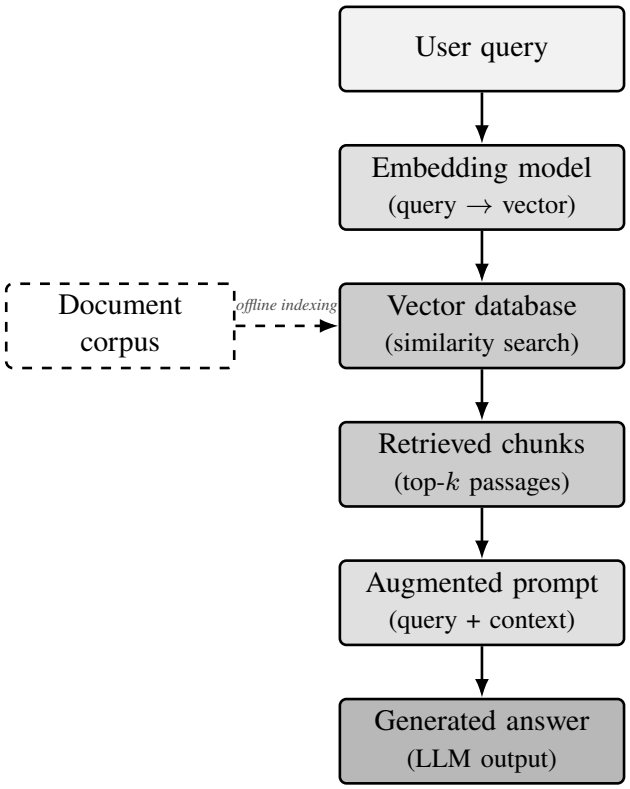


Figure 1. Overview of a Retrieval-Augmented Generation (RAG) pipeline.

2. RELATED WORK

2.1. Financial Question Answering and Financial RAG

Financial question answering presents challenges beyond plain QA because relevant evidence may be distributed across narrative text, numerical values and structured tables. FinQA shows that financial QA can require multi-step reasoning over financial information **ref1**, while TAT-QA shows that questions may require joint reasoning across tables and text, including operations such as addition, comparison and division **ref2**. This makes financial QA sensitive to how evidence is retrieved and presented **ref1ref2**.

These difficulties have encouraged the use of retrieval-augmented approaches, where relevant evidence is retrieved before an answer is generated. Iaroshev et al. (2024) found that financial RAG performance varies across system configurations, question types and document structures, with qualitative questions generally performing better than quantitative ones and text-heavy reports being easier to retrieve from than more complex table- and chart-heavy reports **ref17**. More recent systems have moved towards specialised architectures. FinSage combines multi-modal preprocessing, query decomposition, sparse and dense retrieval, metadata-aware retrieval, HyDE-based query expansion, chunk bundling and domain-specialised reranking **ref16**. Its full pipeline improved retrieval recall from 0.8194 for the FAISS baseline to 0.9251 and achieved stronger FinanceBench QA performance **ref23**.

However, access to more information does not guarantee reliable financial QA. FinanceBench reports hallucination, incorrect answers and latency limitations in evaluated systems **ref19**, while Iaroshev et al. also show that performance depends on model configuration and question type **ref17**. This leaves scope for controlled comparison of retrieval architectures under common generation, data and evaluation conditions.

2.2. Retrieval Architectures for RAG

RAG performance depends strongly on retrieval because generation is conditioned on the evidence supplied to the language model. Retrieval methods differ in how they represent queries and documents and determine relevance. Common approaches include sparse, dense, hybrid, reranking and table-aware retrieval. Broad retrieval benchmarks show that no single method is consistently best across all datasets and domains **ref10ref5ref6**.

2.3. Sparse, dense and hybrid retrieval

BM25 is a sparse lexical retrieval method that ranks documents using the weighted occurrence of query terms, including term frequency, inverse document frequency and document-length normalisation **ref8**. This makes it a relevant baseline for financial QA, where questions may contain exact terminology, company names, accounting labels, numerical values and ticker symbols.

Dense retrieval instead represents queries and passages in continuous embedding spaces, allowing semantic matches without exact lexical overlap **ref6ref7**. Karpukhin et al. **ref6** showed that DPR outperformed a tuned BM25 baseline on four of five open-domain QA datasets, although BM25 performed better on SQuAD where lexical overlap was high. BEIR later showed that retrieval performance varies across domains and tasks, with BM25 remaining a strong zero-shot baseline and dense methods weakening under some domain shifts **ref10**. These findings support evaluating sparse and dense retrieval directly in financial QA rather than assuming results from open-domain benchmarks will transfer unchanged.

Hybrid retrieval combines the complementary characteristics of sparse lexical and dense semantic retrieval. Rankings from different retrievers can be combined using methods such as Reciprocal Rank Fusion, which merges documents according to their positions across retrieval systems **ref9**. BEIR supports this motivation by showing that no single architecture dominates across datasets **ref10**. In finance, FinSage also combines sparse and dense retrieval with metadata and HyDE-based paths, and its full multi-path configuration achieved higher recall than its individual retrieval components **ref16**. However, hybrid retrieval also adds architectural complexity and possible latency, so its benefit in this financial QA setting requires direct evaluation.

2.4. Reranking

The first retrieval identifies candidate passages efficiently, but the most relevant evidence may not always appear at the top. Reranking adds a second relevance stage that re-evaluates retrieved candidates before final context selection. Nogueira et al. **ref12** showed that a T5 reranker improved MRR@10 on MS MARCO from 0.184 for BM25 alone to 0.383, demonstrating that second-stage retrieval ranking can substantially improve an initial retrieval order.

Ranking is also important because language models do not use all context equally. Liu et al. **ref13** found that performance can fall when relevant information appears in the middle of a long context, while adding

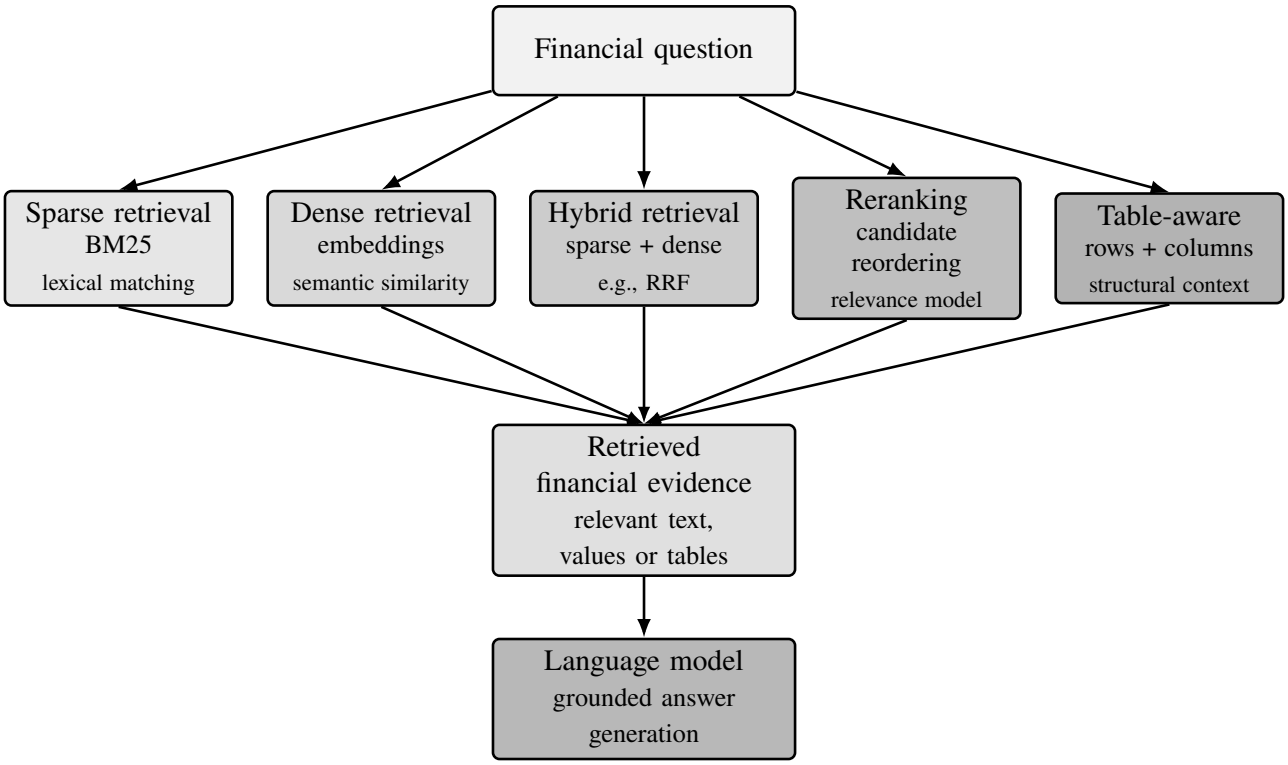


Figure 2. Background taxonomy of retrieval methods relevant to financial QA. Table-aware retrieval is included for context but was not implemented or evaluated in this experiment.

more retrieved documents gave only limited gains once reader performance had saturated. This suggests that selecting and prioritising relevant evidence may be more useful than simply increasing context length.

2.5. Evaluation of RAG systems

RAG evaluation should separate retrieval effectiveness from generation quality. At the retrieval level, metrics such as Recall@k, MRR and nDCG measure whether relevant evidence is retrieved and how highly it is ranked. BEIR uses nDCG@10 as its main cross-dataset retrieval metric **ref10**. This matters because if required evidence is missing from the retrieved context, the generator has no documentary basis for a fully grounded answer.

Generation quality must therefore be evaluated separately. RAGAS measures context relevance, answer faithfulness and answer relevance **ref15**. These dimensions have also been used in financial RAG evaluation by Iaroshev et al. **ref17**. Retrieval effectiveness and generation faithfulness should therefore be treated as related but distinct outcomes.

Financial QA also needs explicit numerical evaluation because a semantically plausible answer can still be financially incorrect. FinanceBench treats incorrect calculations, magnitudes, directions of effect and reasoning that contradicts evidence as incorrect **ref19**. FinQA and TAT-QA also show that financial questions often require numerical operations over structured evidence **ref1**, **ref2**. Numerical correctness is therefore included as an additional measure, especially for quantitative and table-based questions.

Finally, effectiveness must be considered alongside computational efficiency. FinanceBench identifies latency, computational cost and limits in handling large filings as practical issues for long-context approaches **ref23**. BEIR similarly shows that stronger reranking and late-interaction methods can require greater computational cost than simpler retrieval approaches **ref10**, while recent financial RAG work also evaluates latency alongside retrieval quality and faithfulness **ref18**. End-to-end latency is therefore included to assess whether gains from hybrid retrieval, reranking or structured processing justify their additional cost.

Taken together, this literature shows that retrieval architecture, document structure and generation quality can all affect financial RAG performance. However, this reviewed financial RAG studies mainly evaluates complete systems or particular configurations rather than isolating retrieval architecture while keeping the generator, data and evaluation protocol constant. Limited evidence was identified comparing sparse, dense, hybrid, reranked and table-aware retrieval under the same financial QA conditions while jointly measuring retrieval effectiveness, answer faithfulness and latency. This gap motivates the research question and

hypotheses examined in this study.

3. RESEARCH QUESTION AND HYPOTHESES

3.1. Research question

Which RAG architectural design choices produce the best balance between retrieval quality, faithfulness and latency for financial question answering?

3.2. Expected outcomes for P0–P3

- **P0:** A closed-book LLM will exhibit the highest hallucination rate and lowest answer accuracy, but will achieve the lowest latency.
- **P1:** Dense RAG will improve faithfulness over closed-book answering, but will struggle with exact financial terminology, tickers and numerical references.
- **P2:** Hybrid retrieval will outperform dense-only retrieval on retrieval recall and numeric/terminology-heavy questions.
- **P3:** Hybrid retrieval with reranking will produce the highest faithfulness among text-based pipelines, but with a measurable latency cost.

4. METHODOLOGY

4.1. Dataset

4.1.1. Data Source

The financial question-answering data used in this study is FinanceBench **ref23** . FinanceBench serves as the benchmark for evaluating document-level financial question answering and retrieval performance in this paper. The use of this benchmark follows previous financial RAG research in which retrieval architectures have been evaluated across FinanceBench.
FinanceBench is an open-book financial question-answering benchmark constructed from public financial documents of publicly traded companies **ref23**. Although the complete FinanceBench dataset contains more than 10,000 question-answer-evidence triplets, the publicly available evaluation subset contains 150 manually curated cases. These 150 cases were used in full rather than constructing new question-answer pairs. The subset contains 50 domain-relevant questions, 50 novel-generated questions, and 50 metrics-generated questions, providing variation in information extraction, numerical reasoning, and logical reasoning tasks.
Each FinanceBench instance contains a question, reference answer, supporting evidence, source document information and, where applicable, a justification for the answer. The corresponding financial documents referenced by the 150 questions were obtained from the FinanceBench document repository and used to construct the retrieval corpus. These documents include financial filings and reports such as 10-K, 10-Q, 8-K and earnings-related documents. Only documents corresponding to the benchmark questions were included in the experimental corpus, ensuring alignment between each question, its reference answer and the source information from which that answer was derived.
Rather than constructing a separate index for each question, all unique FinanceBench source documents associated with the evaluation set were combined into a shared retrieval corpus. This produces a more realistic retrieval task in which the system must identify relevant evidence from multiple financial documents rather than being given the correct document in advance.

4.1.2. Question-answer pairs

No new question-answer pairs were created for this study. The original questions, reference answers and evidence annotations supplied by FinanceBench. This decision reflects the primary objective of the study, which is to compare RAG architectural design choices rather than to construct or validate a new financial QA benchmark.

4.2. Compared Architecture

Four QA configurations were compared: the closed-book baseline P0 and three RAG pipelines P1–P3. They share the frozen question set and answer model, while differing in retrieval method, candidate depth, and reranking. Consequently, comparisons estimate differences between the complete configurations specified in Table 1; they are not equal-depth ablations of BM25 fusion or reranking.

4.3. Controlled Variable

The public FinanceBench sample contains 150 questions. Five specified question IDs were used for development smoke tests and excluded from the primary comparison. The remaining 145 questions were evaluated under every configuration, yielding 580 saved answers. The split script preserved source question order and recorded question IDs and file hashes in a manifest, allowing the disjoint split and evaluation inputs to be checked.

4.4. Implementation Details

The four pipelines were implemented over the frozen 24,410-chunk document corpus. P0 answered without

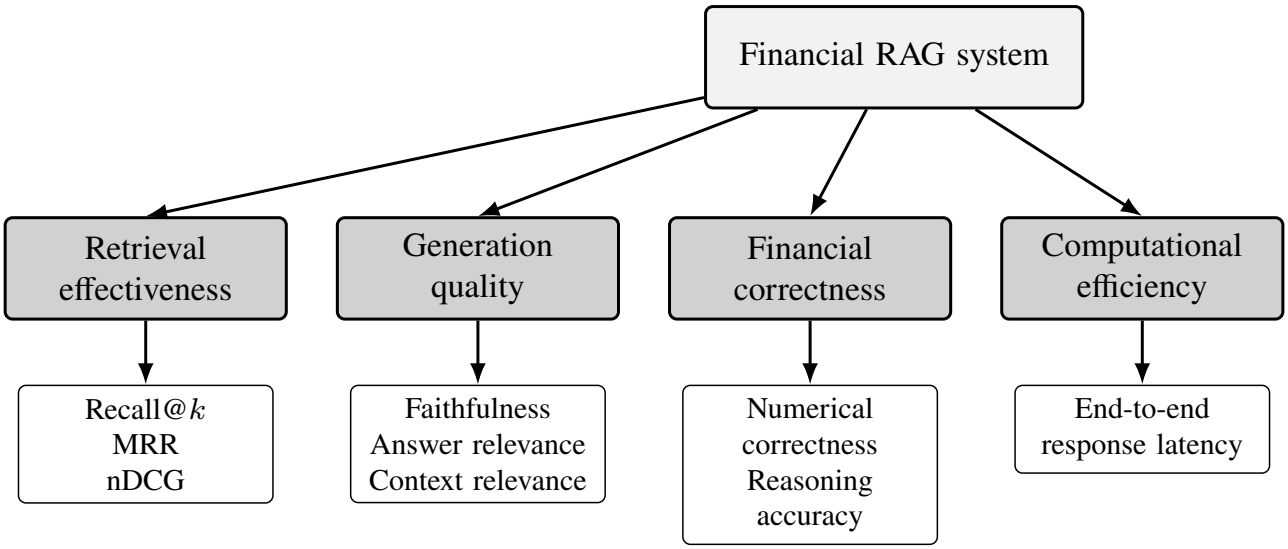


Figure 3. Evaluation dimensions for financial RAG systems. Retrieval effectiveness measures whether relevant evidence is retrieved and ranked appropriately, while generation quality assesses the relevance and faithfulness of generated answers. Numerical correctness captures financial and quantitative accuracy, and latency measures computational efficiency.

Table 1. Experimental RAG architectures and their expected trade-offs.

ID	Architecture	Description	Trade-off
P0	Closed-book LLM	No retrieval	Lowest complexity and latency; no retrieved evidence for grounding
P1	Dense RAG	Dense retrieval only	Captures semantic similarity; may miss exact lexical matches
P2	Hybrid RAG	BM25 + dense fusion	Combines lexical and semantic evidence; increases retrieval complexity
P3	Hybrid + Reranking	Cross-encoder reranking	Improves candidate prioritisation; adds computational cost and latency

Table 2. Common experimental settings.

Variable	Setting
Evaluation questions	Same 145 held-out FinanceBench questions
Question order	Identical for P0–P3
Answer model	`gpt-4o-mini`
Maximum output tokens	256
Temperature	Unset; API default
Document corpus	Frozen, 24,410 chunks
RAG prompt builder	Identical for P1–P3
Query preprocessing	Identical for P1–P3
Company and year filters	Identical for P1–P3
Final answer context	Five chunks for P1–P3, when available

retrieval. P1 selected the five highest-ranked chunks using E5 dense retrieval. P2 combined the top 10 E5 and top 10 BM25 results using reciprocal rank fusion (RRF; $k = 60$), then selected five chunks. P3 fused the top 20 results from each retriever, reranked the 20 highest-ranked fused candidates with `cross-encoder/ms-marco-MiniLM-L6-v2` and selected five chunks. All four pipelines used `gpt-4o-mini` to generate answers.

4.4.1. Reproducibility Details

The source PDFs were processed using `pypdf` text extraction without OCR. All physical pages, including those without extractable text, retained their zero-indexed positions. The extracted text was divided into page-local token windows using the `intfloat/e5-base-v2` tokenizer. The maximum embedding-model input was 512 tokens, including reserved space for the prefix and special tokens, with a 50-token content overlap. Chunks never crossed page boundaries. This produced the frozen corpus of 24,410 chunks.

The five development question IDs were specified in `configs/question_split.json`. The ordered evaluation IDs and file hashes were recorded in the split manifest. For retrieval evaluation, a chunk matched annotated FinanceBench evidence when both its document identifier and zero-indexed page number matched an evidence reference. A stricter mapping

Table 3. Implemented architectures.

ID	Candidate selection	Final context
P0	Closed-book; no retrieval	None
P1	E5 dense top 5	Top 5
P2	Dense top 10 + BM25 top 10; RRF	Fused top 5
P3	Dense top 20 + BM25 top 20; RRF + reranking	Reranked top 5

Table 4. Index and reranking settings.

Item	Setting
Frozen corpus file	`data/processed/chunks.jsonl`
Dense index	FAISS
Sparse index	BM25
Fusion method	Reciprocal rank fusion (RRF)
RRF rank constant	$k = 60$
P3 fused candidate count	20
P3 reranker	`cross-encoder/ms-marco-MiniLM-L6-v2`
Reranker pair limit	512 tokens
Long-candidate handling	Overlapping windows; 64-token overlap

from evidence text to individual chunks was retained separately but was not used in place of the page-level retrieval labels.

5. EVALUATION PROTOCOL

The experiment contains 145 held-out questions and four architectures ($145 \times 4 = 580$ saved answers). Each question was scored separately and reported means are averages of the applicable question-level scores. The number of applicable questions is reported where it differs from 145. Lastly NA indicates that a metric could not be scored rather than a score of zero.

5.1. Retrieval Evaluation

For P1–P3 the five chunks supplied to the answer model were compared with FinanceBench's annotated evidence pages. A match required the same document and zero-indexed page. Multiple chunks from one gold page counted once for recall while empty retrievals received zero and remained in the 145-question denominator. These page-level metrics measure whether annotated evidence was available in the final context and they do not establish that a chunk contained the precise answer-bearing passage.

5.2. Answer quality Evaluation

Generated answers were compared with FinanceBench reference answers using normalized exact match, token F1, numeric-entity measures, model-judged whole-answer correctness, reference-unit coverage and BERTScore. Factual-unit support was assessed separately against the final retrieved chunks and annotated FinanceBench evidence.. Of the 145 questions there were 112 questions whose reference answers contain an extractable finance number for the numeric summary means.

Table 5. Retrieval metric definitions.

Metric	Definition
Hit@k	Indicator that at least one gold evidence page occurs among the first k retrieved chunks.
Recall@5	Number of distinct gold pages represented in the first five chunks divided by the number of gold pages.
MRR@5	Reciprocal rank of the first gold page in the top five; zero if absent.

P0 has no retrieved context, so its context-support metrics are NA. All four pipelines used gpt-4o-mini for answer generation. Evidence support was subsequently assessed in separate calls to the same model(gpt-4o-mini) using the fixed answer-unit-passage-v4 judge protocol. Because the generator and judge used the same model these support scores are automated judgments rather than independent human assessments.

5.3. Latency Evaluation

The retrieval, reranking and answer-generation time taken were recorded for each question where applicable. Total latency is the sum of those recorded stages which was reported in milliseconds and averaged across questions. Timing excludes pipeline and model initialization, and individual measurements may be affected by machine load, caching, and API response

Table 6. Answer and evidence-support metric definitions.

Metric	Definition
Exact match (EM)	Indicator of equality after lowercasing and removing ASCII punctuation and articles.
Token F1	Harmonic mean of precision and recall over normalized token occurrences.
Numeric accuracy	Matched gold numeric entities divided by gold numeric entities.
Numeric precision	Matched numeric entities divided by generated numeric entities.
Numeric F1	Twice the matched numeric-entity count divided by the sum of gold and generated counts.
Context support	Supported factual answer units divided by scored factual units, using the final retrieved chunks.
Gold support	Supported factual answer units divided by scored factual units, using annotated benchmark evidence.
Hallucination rate	$1 -$ support for the corresponding evidence source.
Exact abstentions	Count of answers matching the evaluation's exact abstention criterion.
Failure to answer	Outcome assigned to an exact abstention or judged for another response supplying no substantive requested answer.
Unresolved judge errors	Evaluation outputs remaining invalid or inconsistent after three attempts. Their semantic scores remain missing.
Whole-answer correctness	Binary model judgment: one when every required reference-rubric unit is correctly answered without a substantive contradiction; zero for other valid outcomes. Exact abstentions receive zero deterministically.
Reference-unit coverage (completeness proxy)	Reference units judged correctly answered divided by the total units in the shared rubric.
BERTScore F1	Contextual token similarity between generated and reference answers. Calculated using BERT-base, layer 9, without IDF weighting or baseline rescaling. It measures similarity rather than factual correctness.

variation. No per-answer monetary cost was measured.

Table 7. Timing fields and measurement scope.

Field	Recorded interval
Retrieval latency	Retrieval and selection time, where applicable.
Reranking latency	Cross-encoder reranking time for P3.
Generation latency	Answer-model response time.
Total latency	Per-question retrieval, reranking and generation time, as applicable.
Cost	No per-answer monetary cost measurement is reported.

6. RESULTS AND ANALYSIS

6.1. Evaluation Scope

The four pipeline configurations were evaluated on the same 145 held-out FInanceBench questions. P0 generated answers without retrieval so retrieval metrics only applies to P1–P3. The abstention analysis and the answer correctness, reference coverage and BERTscore evaluation were performed on the saved answers. The latency and retrieval conetexts were also evaluated.

The evaluation distinguishes retrieval of annotated pages, support for generated factual units, whole-answer correctness, coverage of reference answer units, textual similarity, and exact abstention. These measures have different denominators and meanings. A response can be supported by its retrieved context while failing to answer the question correctly; a response can also avoid the prescribed abstention phrase while providing no substantive answer.

6.2. Main Result

6.2.1. Retrieval of Annotated Evidence

Table 8 shows that P3 placed an annotated evidence page first most often with Hit@1 of 0.214 and MRR@5 of 0.284. However P1 and P3 had identical Hit@5 (0.4) while their Recall@5 values only differed by 0.007. P2 scored below P1 on every reported metric.

6.2.2. Whole Answer Correctness and Reference Coverage

A post-hoc evaluation used `gpt-4o-mini` to generate 145 reference-unit rubrics from the questions, reference answers and annotated FinanceBench evidence, without seeing any pipeline answers. One rubric per question was shared across P0–P3. The 179 exact abstentions were assigned deterministic failure-to-answer outcomes and zero correctness and coverage scores. The remaining 401 responses were submitted to `gpt-4o-mini` for comparison with the question, reference answer, annotated evidence and

Table 8. Retrieval of annotated evidence pages by the three RAG configurations. Scores are question-level means over all 145 evaluation questions, based on the final ranked chunks supplied to the answer model. "Empty" counts questions with no retrieved chunks; these questions received zero retrieval scores.

ID	Hit@1	Hit@3	Hit@5	Recall@5	MRR@5	Empty
P1	0.110	0.303	0.400	0.362	0.221	2
P2	0.097	0.234	0.317	0.290	0.174	2
P3	0.214	0.352	0.400	0.369	0.284	2

shared rubric. The judge did not receive the pipeline identity or the retrieved context.

Whole-answer correctness was binary: a valid judgment scored one only if it addressed every required rubric unit without any substantive contradiction; all other valid outcomes scored zero. Partial answers were recorded separately and were not treated as fully correct.

The additional completeness measure was the proportion of generated reference units judged correctly answered, macro-averaged across questions, and is reported as a *reference-unit coverage proxy*. Although the rubrics were intended to include only essential requested information, inspection showed that some also required optional supporting quantities. Consequently, this measure may not reflect only the minimum content needed to answer a question. An answer can receive partial coverage credit while still being incorrect if it gives a wrong requested result or contains contradictory reasoning.

Of the 401 responses submitted to the judge, 390 produced valid judgments. Together with the 179 deterministic abstention scores, these formed 569 valid semantic results across the 580 responses. The remaining 11 judgments failed consistency validation after three attempts and were retained as unresolved judge errors, with no imputed semantic score. Table 10 gives the full outcome counts. P1 and P3 each had 38 answers judged fully correct, compared with 31 for P2 and 14 for P0. P1's slightly higher valid-judgment correctness mean than P3's arises from the different denominators, 142 and 143 rather than an additional correct answer.

The uncertainty introduced specifically by unresolved judgments can be made explicit over the full 145-question denominator. If c is the number judged correct and u is the number unresolved, the lower and upper correctness bounds are $c/145$ and $(c+u)/145$. These bounds are 0.097–0.110 for P0, 0.262–0.283 for P1, 0.214–0.241 for P2, and 0.262–0.276 for P3. They are not confidence intervals and do not account for undetected judge mistakes.

Because the RAG configurations abstained on different questions, the conditional results are also reported in Table 11. These exclude exact abstentions; semantic means additionally exclude unresolved judge errors. A longer procedural response or noncanonical refusal remains in this non-exact abstention population, including when it is judged a failure to answer.

Conditional correctness was 0.452 for P1 and 0.447 for P3, compared with 0.397 for P2. These means describe selected responses, rather than performance over all 145 questions. They should be read alongside abstention and scoring coverage, rather than used alone to rank the configurations.

The `bert-base-uncased` layer 9 for the BERTScore, without IDF weighting or baseline rescaling. All 580 answer–reference pairs were scored without truncation. Its F1 score here measures the contextual token similarity. P3 had the highest overall BERTScore F1 0.476, followed closely by P0 at 0.475, this happened despite P0 and P3 having a large difference in correctness score so it can be determined that the similarity and correctness yield different descriptions of these responses. BERTScore values also depend on the encoder and scoring configuration and should not be directly compared with results using another variant.

6.2.3. Lexical and Numeric Diagnostics

The three RAG pipelines performed better P0 on numeric accuracy with P2 scoring the highest (0.190). Exact match (EM) remained low across all recorded systems with P1 matching two reference answers exactly while the others matched only one. Token F1 and exact match measure textual overlap should not be interpreted as rates of fully correct answers. EM and token F1 use $n = 145$ per architecture. Numeric columns use $n = 112$ per architecture.

6.2.4. Factual-Unit Support and Hallucination Proxies

For each answer with at least one judge-scored factual unit, support was the proportion of those units supported by the selected evidence source. The corresponding hallucination proxy was defined as $1 -$ support, the proportion of scored units judged unsupported. The reported values are question-level macro means over the same applicable answers for each support–hallucination pair. Consequently, each pair

Table 9. Post-hoc model-judged whole-answer correctness and reference-unit coverage against the question, reference answer and annotated gold evidence, with BERTScore similarity to the reference answer. Correctness and coverage are means over the valid semantic judgments shown in n; exact abstentions are valid zero scores. Judge errors have missing semantic scores. BERTScore F1 uses all 145 answers per configuration, including abstentions.

ID	Valid n	Judge errors	Correctness	Coverage proxy	BERTScore F1
P0	143	2	0.098	0.135	0.475
P1	142	3	0.268	0.299	0.467
P2	141	4	0.220	0.256	0.466
P3	143	2	0.266	0.300	0.476

Table 10. Whole-answer outcome counts from the post-hoc reference-based evaluation. The five outcome columns partition all 145 responses per configuration. Failure to answer includes prescribed exact abstentions and broader responses judged not to supply a substantive answer. An unresolved judge error is a scoring failure, not an incorrect-answer judgment.

ID	Correct	Partially correct	Incorrect	Failure to answer	Judge error
P0	14	8	114	7	2
P1	38	6	40	58	3
P2	31	7	39	64	4
P3	38	5	42	58	2

Table 11. Answer evaluation conditional on not producing a prescribed exact abstention. "Responses" counts the entire non-exact-abstention subset; "Valid semantic n" excludes unresolved judge errors. Correctness and reference-unit coverage use the latter denominator. BERTScore uses all responses in this subset. The selected question sets differ across pipelines.

ID	Responses	Valid semantic n	Correctness	Coverage proxy	BERTScore F1
P0	145	143	0.098	0.135	0.475
P1	87	84	0.452	0.506	0.557
P2	82	78	0.397	0.463	0.568
P3	87	85	0.447	0.505	0.572

Table 12. Mean answer-comparison scores by configuration. EM and token F1 use all 145 questions; numeric scores use the 112 questions with extractable reference numbers and the corrected v2 evaluation.

ID	EM	Token F1	Numeric accuracy	Numeric precision	Numeric F1
P0	0.007	0.172	0.056	0.051	0.048
P1	0.014	0.159	0.182	0.123	0.130
P2	0.007	0.166	0.190	0.130	0.137
P3	0.007	0.170	0.176	0.114	0.127

Table 13. Model-judged factual-unit support and hallucination proxies. Context refers to the final retrieved chunks; gold refers to annotated FinanceBench evidence. Parentheses give the number of scored answers.

ID	Abstentions	Context support	Context halluc.	Gold support	Gold halluc.
P0	0	NA	NA	0.583 (112)	0.417
P1	58	0.850 (85)	0.150	0.636 (83)	0.364
P2	63	0.820 (80)	0.180	0.621 (81)	0.379
P3	58	0.832 (86)	0.168	0.652 (85)	0.348

sums to one: for example, P1's context support of 0.850 corresponds to a context hallucination proxy of 0.150. Context and gold scores use different evidence sources and should not be added to each other.

P1 had the highest conditional context-support mean (0.850), while P3 had the highest conditional gold-support mean (0.652). Answers without scored factual units were excluded, and the applicable sets differ across pipelines and evidence sources. P1, P2 and P3 also abstained on 58, 63 and 58 questions, respectively. These scores must therefore be interpreted alongside their scored denominators and abstention counts. P0 has no context score because it received no retrieved chunks.

6.2.5. Latency

P1 had the lowest observed mean total latency at 1,933ms followed by P0 at 2,074 then P2 at 2,246ms. P3's latency averaged at 5,633ms of which 3,366ms was recorded as reranking time. These measurements

Table 14. Per-question response latency in milliseconds ($n = 145$ per configuration). Total time includes retrieval, reranking and answer generation where applicable; the median and standard deviation (SD) describe total time. NA marks stages a configuration does not use.

ID	Total mean	Median	Total SD	Retrieval mean	Rerank mean	Generation mean
P0	2,074	1,916	1,006	NA	NA	2,074
P1	1,933	1,795	853	86	NA	1,847
P2	2,246	2,176	777	461	NA	1,785
P3	5,633	5,619	903	466	3,366	1,800

exclude pipeline and model initialization.

6.3. Abstention Behaviour

6.3.1. Frequency, Shared Questions and Overlap

From P1–P3 there were 179 exact abstention among the 435 RAG pipeline responses (41.1%) . These were spread across 77 distinct questions (53.1% of the 145). P1 and P3 each abstained on 58/145 questions (40.0%), leaving 87 responses without an exact abstention. P2 abstained on 63/145 (43.4%), leaving 82. P0 recorded no exact abstentions. Including P0 gives 179/580 responses (30.9%), although the RAG-only rate is more directly relevant to refusal under the retrieved-context restriction.

The frozen exact-abstention detector checks whether the entire normalised response matches one of five prescribed refusal phrases. All 179 detected responses used "Insufficient evidence in the retrieved context." Longer refusals, procedural explanations and incomplete answers can remain outside this count. Table 10 demonstrates the distinction: P0 had seven model-judged failures to answer despite zero exact abstentions, and P2 had 64 failures to answer compared with 63 exact abstentions. These broader labels are themselves subject to judge error.

Table 15. Overlap of exact-abstention decisions across the three RAG configurations on the same 145 questions. "Only" excludes abstention by the remaining configuration or configurations.

Abstaining configurations	Questions	Percentage
None	68	46.9%
P1 only	4	2.8%
P2 only	7	4.8%
P3 only	7	4.8%
P1 and P2 only	8	5.5%
P1 and P3 only	3	2.1%
P2 and P3 only	5	3.4%
P1, P2 and P3	43	29.7%
Total	145	100.0%

When looking at the common abstention behaviour. 43 happened across P1–P3 which represents 55.8% of the 77 questions. These shared cases accounted for 74.1% of P1's abstentions, 68.3% of P2's and 74.1% of P3's. No pipeline exact-abstained on 68 questions; the remaining 34 had mixed decisions. Exactly one pipeline abstained on 18 questions and exactly two on 16.

6.3.2. Evidence Availability and Section Stages

Among abstentions, a final annotated gold-page hit was absent on 49/58 P1 responses, 57/63 P2 responses and 48/58 P3 responses. Together, these account for 154/179 abstention decisions (86.0%). Table 16 partitions the exact abstentions by the observed retrieval stage at which annotated-page coverage was absent or lost.

P1's dense top five formed its final context. P2 combined dense and sparse top-ten results before selecting the RRF top five. P3 combined dense and sparse top-twenty results, retained the RRF top twenty and selected the reranked top five. Consequently, these diagnostics describe complete configurations with different search budgets. They do not isolate the effect of reranking alone. Each pipeline returned an empty context on the same two questions; most refusals therefore occurred with some retrieved material available. Empty retrieval is included in Table 16 and must not be added to its totals.

The abstentions happened more frequently without a final gold-page hit. Conditional on having no hit, the rates were 49/87 (56.3%) for P1, 57/99 (57.6%) for P2 and 48/87 (55.2%) for P3. With a hit, the rates were 9/58 (15.5%), 6/46 (13.0%) and 10/58 (17.2%), respectively. These denominators contain all questions with the relevant retrieval condition, rather than only abstentions. They show an association between annotated evidence availability and refusal, without establishing an individual model's internal reason.

Judging from the 43 commonly abstained questions from P1–P3, 35 had no final gold page hit in any pipeline while three had hits in all 3 pipelines and fuve had mixed retrieval outcomes. Also noting, 22 of the 68 questions without any exact abstention had no final gold page hit in any pipeline and neither condition determined answer correctness: unannotated passages may provide valid alternative evidence, while a chunk

Table 16. Observed annotated-page retrieval conditions among exact abstentions. These mutually exclusive stage conditions partition the 179 abstaining RAG responses; they are not validated internal reasons for refusal. Candidate-pool absence includes the two empty retrievals per configuration.

Observed condition	P1	P2	P3	Total
Gold page absent from observed candidate pool	49	40	29	118
Gold page present in pool but lost at RRF cutoff	0	17	7	24
Gold page retained before reranking but absent from final five	0	0	12	12
At least one gold page present in final context	9	6	10	25
Total exact abstentions	58	63	58	179

from an annotated page may omit the decisive passage.

6.3.3. Abstention Despite Retrieval of an Annotated Page

There were 25 exact abstention even when there was at least one annotated gold page being present in the final context, affecting 16 distinct questions. These comprised of nine P1 decisions, six P2 decisions and ten P3 decisions which represents 15.5%, 9.5% and 17.2% of each pipeline's abstentions, respectively, or 14.0% of all 179 abstentions. These percentages differ from the conditional rates above because the denominators here are abstentions.

Ten of the 25 decisions, across six questions, retrieved only one of two annotated evidence pages. The remaining 15 decisions concerned questions with a single annotated page, which was retrieved. This verifies page coverage, rather than sufficient answer-bearing passage coverage.

An evidence-selection example helps explain differences across pipelines. For Best Buy's FY2019 inventories (`financebench_id_04417`), P1 and P2 abstained, while P3 answered "$5,409 million," matching the gold value. P1's observed candidate pool lacked the annotated page; P2 found an annotated-page candidate but lost it at the RRF cutoff; P3 retained a final chunk explicitly listing merchandise inventories of 5,409 million. This is an observed correspondence between evidence surviving selection and the response decision, without establishing what would happen under a controlled change to one component.

6.3.4. Question Types and Broader Failures to Answer

Metric generated questions produced the highest abstention rate in P1–P3 while Novel generated had the lowest. The numerical reasoning labelled questions had their rates as 22/43 (51.2%), 23/43 (53.5%) and 25/43 (58.1%) for P1–P3. For pure Information extraction, they were 7/29 (24.1%), 8/29 (27.6%) and 5/29 (17.2%). These differences are consistent with difficulty obtaining and combining numerical inputs, but the labels do not establish a cause. Mixed reasoning labels form separate categories, and 49 questions lacked reasoning annotations.

Table 17. Exact abstentions by native FinanceBench question type. Each percentage uses the number of questions in its row. These are descriptive subgroup results, not additional hypothesis tests.

Question type	n	P1	P2	P3
Domain-relevant	48	19 (39.6%)	19 (39.6%)	17 (35.4%)
Metrics-generated	48	24 (50.0%)	28 (58.3%)	26 (54.2%)
Novel-generated	49	15 (30.6%)	16 (32.7%)	15 (30.6%)

6.4. Paired Comparison

Paired comparisons aligned answers by FinanceBench question ID and calculated each difference as configuration B minus A. Binary outcomes, including Hit@5 and exact match, were tested with two-sided exact McNemar tests. Graded scores and latency used two-sided Wilcoxon signed-rank tests, excluding zero differences from the rank calculation. Missing or inapplicable scores were excluded pairwise, rather than imputed as zero. Seeded paired bootstrap resampling with 10,000 replicates produced 95 percentile intervals for mean paired differences. Holm adjustment was applied across all 21 specified contrasts. These contrasts were specified after earlier aggregate results had been inspected. The inferential analysis should therefore be read as exploratory, rather than as a blinded preregistered test. Moreover, the bootstrap intervals estimate mean differences whereas Wilcoxon tests signed ranks; their thresholds need not agree. For P2 versus P3, the paired context-support difference was $+0.013$ (95% bootstrap interval $[-0.065,\ 0.091]$; Holm-adjusted $p = 1.000$). The mean total-latency difference was $+3{,}387$ ms (95% bootstrap interval $[3{,}216,\ 3{,}554]$ ms; Holm-adjusted $p < 0.001$). The latter establishes a substantial latency difference between these complete configurations; the former does not establish a faithfulness improvement.

In these paired tests, the detected improvements in the numeric accuracy for P1 and P3 is relative to P0 but did not detect a gold-support gain for P1 over P0. P2's

Table 18. Selected paired comparisons. Mean differences are calculated as B minus A; positive latency differences indicate slower responses. Pairs are questions with valid scores for both configurations. Reported p-values are Holm-adjusted across 21 comparisons.

A → B	Metric	Mean difference	Pairs	Holm p
P0 → P1	Gold support	+0.083	66	1.000
P0 → P1	Numeric accuracy	+0.126	112	0.030
P0 → P3	Numeric accuracy	+0.121	112	0.028
P1 → P2	Hit@5	−0.083	145	0.177
P1 → P2	Recall@5	−0.072	145	0.184
P2 → P3	Context support	+0.013	71	1.000
P2 → P3	Total latency (ms)	+3,387	145	< 0.001

observed retrieval scores were lower than P1's for the listed Hit@5 and Recall@5 but the differences were not significant after correction. P3's context-support gain over P2 was small and not significant, while its mean total latency increased by 3,387 ms ($p < 0.001$). Thus, the experiment establishes a **substantial latency cost** for the complete P3 configuration without establishing the predicted faithfulness gain.

P3's stronger first rank retrieval did not produce a corresponding improvement across the measured answer metrics. Retrieving an annotated page does not guarantee that the selected chunk contains the answer or that the llm generator will use the correct year, value or calculation. Also it is worth noting that P3 searches a wider candidate pool than p@ before reranking so their difference cannot be attributed solely to the cross-encoder.

7. DISCUSSION

The answer evaluations encompasses beyond word and number matching but does not independently verify correctness. P0 had the lowest model-judged correctness and reference-unit coverage; P1 and P3 each achieved 38 fully correct answers, while P2 achieved 31. This supports a possible retrieval benefit in this run, but the metrics were assessed post hoc using an imperfect automated judge. The whole answer correctness was model assessed and not validated by independent human review.

Despite P2's highest numeric-entity recall did not lead to the highest semantic correctness or coverage and while P3 had the highest BERTScore it did not yeild more fully correct answers than P1. These findings therefore show that numeric matching, reference similarity, factual-unit support and whole answer correctness must be reported separately. A hallucination proxy based on unsupported factual units is not a verified false-answer rate not can its complementary support score be treated as accuracy.

Abstention was a major behaviour in the experiments, over 40% of RAG responses were exact refusals and most abstentions in the 3 pipelines came from the same 43 questions abstained by all three. Most common was the missing annotated evidence which was linked to higher refusal rates. However, pipeline-specific decisions affected 34 questions, and identical P1/P3 abstention totals hid 24 changed decisions. The pipelines therefore differed not only in response quality but also in which questions they attempted; conditional quality averages alone cannot capture this trade-off.

Another point worth noting was the 25 refusals despite annotated page retrieval which shows that locating relevant page does not guarantee sufficient usable evidence. Some contexts included only part of a multi-page calculation or incomplete table fragments, while AMD showed a case where explicit answer-bearing evidence was available yet the generator still refused. Possible improvements include retaining related table fragments, preserving multi-page inputs, and reducing irrelevant context which were not tested. Because refusal text does not reveal the model's internal reason, the observed conditions should not be treated as confirmed explanations for every refusal.

The prediction that P0 would be the fastest was not supported as P1 had the lowest latency. P0 also had the highest descriptive gold-evidence hallucination proxy, but the paired gold-support tests did not establish an inferential claim. Likewise, P1's proposed gold-support advantage over P0 and P3's proposed context-support advantage over P2 were not established. Non-significant results do not demonstrate that configurations perform equally.

P3 performed better in first rank retrieval but with a significant higher latency and no more fully correct answers than P1. P1 and P3 also matched on Hit@5 and exact abstention totals. Due to this P1 is therefore a provisional latency conscious baseline and not necessarily the most factually correct architecture. Because P3 searched a larger candidate pool than P2 before reranking, the P2–P3 comparison cannot isolate the cross-encoder's causal effect. Similarly, the native question-type and reasoning findings are

descriptive and do not confirm the earlier predictions about terminology-heavy or numeric-heavy subgroups.

8. LIMITATION AND CONCLUSION

Several limitations affect interpretation. Retrieval was evaluated against annotated pages rather than specific passages or table cells, potentially overlooking valid evidence or incomplete coverage. Abstention detection was limited by narrow exact-match rules and model-judged classifications, while qualitative examples were illustrative rather than exhaustive.

Support and correctness were assessed using gpt-4o-mini (the same model as the generator) without independent human validation. Inspection revealed rubric omissions, overly strict judgments, and inconsistent treatment of valid answers. Shared rubrics could propagate these errors, while correctness bounds addressed only 11 unresolved judgments, not undetected errors. EM, F1, numeric-entity scores, BERTScore, and reference-unit coverage remain complementary proxies rather than definitive correctness measures. BERTScore settings and methodological differences also limit cross-study comparisons.

The study involved a single 145-question run without repeated trials, tail-latency measurements, or cost analysis. Timings excluded startup and evaluation, while unequal retrieval depths prevented isolating reranking effects. Statistical tests were exploratory, and additional analyses descriptive.

P3 improved evidence ranking but increased latency without established correctness or context-support gains over P1, while P2 did not improve mean retrieval performance. Abstentions reflected both evidence-access limitations and configuration-dependent refusals, sometimes despite sufficient context. P1 remains a provisional latency-conscious baseline, not an accuracy optimum. Future work should validate judgments independently, assess passage-level evidence, standardise retrieval depths, and evaluate answer coverage alongside quality.

9. TERMINOLOGIES

Table 19. Evaluation metrics and aggregation terminology.

Term	Meaning in this study
EM	Exact match between the normalised generated and reference answers.
F1	Harmonic mean of precision and recall.
Token F1	F1 based on token overlap between generated and reference answers.
Precision	Proportion of predicted items matched to the reference under the metric's matching rule.
Recall	Proportion of reference items matched by the prediction under the metric's matching rule.
Hit@k	Whether at least one annotated evidence page occurs within the first k retrieved results.
Recall@5	Proportion of annotated evidence pages covered by the final five retrieved chunks.
MRR@5	Mean reciprocal rank of the first annotated-page hit within the final five results; zero when none is found.
Numeric accuracy	Question-level recall of reference numeric entities; a numeric-matching proxy.
BERTScore F1	Contextual token-similarity F1 between generated and reference answers; here, BERT-base layer 9 without baseline rescaling.
Macro mean	Unweighted mean of per-question scores over the applicable questions.
Conditional mean	Mean over a stated subset, such as valid scored responses without an exact abstention.

Table 20. Statistical terminology, scoring denominators and units.

Term	Meaning in this study
SD	Standard deviation.
n	Number of observations in the stated applicable scoring population.
NA	Not applicable or unavailable; unresolved judge errors are reported separately.
CI	Confidence interval; here, a paired-bootstrap 95% percentile interval for a mean difference.
Paired comparison	Comparison of configurations on matched question IDs.
Paired bootstrap	Resampling matched question pairs with replacement to estimate uncertainty in a paired statistic.
McNemar test	Test comparing paired binary outcomes; the study uses its two-sided exact form.
Wilcoxon signed-rank test	Paired test based on the signed ranks of nonzero score differences.
p-value	Probability, under the null hypothesis and test assumptions, of a result at least as extreme as observed.
Holm p	p-value adjusted for multiple comparisons using Holm's method.
ms	Milliseconds; the unit used to report response latency.

Table 21. Answer correctness, reference coverage and factual-support terminology.

Term	Meaning in this study
Reference answer / gold answer	Annotated FinanceBench answer used as the comparison reference.
Gold evidence	Annotated FinanceBench evidence supporting the reference answer.
Generated factual unit	A factual claim extracted from a generated answer for evidence-support assessment.
Reference unit	Component of a model-generated reference rubric, intended to represent required answer content.
Reference rubric	A fixed set of reference units generated for one question and shared across all four configurations.
Whole-answer correctness	Binary model judgment that all required rubric units are answered without a substantive contradiction. Exact abstentions receive zero deterministically.
Reference-unit coverage proxy	Fraction of generated reference units judged correctly answered; the reported completeness proxy.
Context support	Fraction of scored generated factual units judged supported by the final retrieved chunks.
Gold support	Fraction of scored generated factual units judged supported by annotated FinanceBench evidence.
Hallucination proxy	One minus the corresponding support score: the fraction of scored units judged unsupported, rather than independently verified false.
Partially correct	A substantive answer covering at least one required reference unit, with others missing and no incorrect required unit or substantive contradiction.
Valid semantic result	A deterministic exact-abstention score or an accepted model judgment passing the evaluation's consistency checks.

Table 22. Abstention, scoring coverage and unresolved-judgment terminology.

Term	Meaning in this study
Exact abstention	Entire normalised response matches one of the prescribed refusal phrases.
Non-exact abstention	A response outside the exact-abstention detector; it can still be incomplete, incorrect or a broader refusal.
Failure to answer	Outcome assigned to an exact abstention or judged for another response supplying no substantive requested answer.
Judge error	Unresolved invalid or inconsistent evaluation output after the retry limit; semantic scores remain missing.
Scoring coverage	Number or proportion of responses with a valid score for a particular metric; distinct from answer-content coverage.
Correctness bounds	Full-question lower and upper rates obtained by assigning unresolved judgments zero and one, respectively. These are not confidence intervals.
Jaccard overlap	Number of shared abstentions divided by the number of questions on which either configuration abstained.
Gold-page hit	At least one retrieved chunk comes from an annotated evidence page; this does not establish sufficient answer-bearing passage coverage.

Table 23. System and implementation terminology.

Term	Full meaning or role in this study
QA	Question answering.
RAG	Retrieval-augmented generation.
LLM	Large language model.
BM25	Best Matching 25; the sparse retrieval method.
RRF	Reciprocal rank fusion; combines ranked retrieval lists.
FAISS	Facebook AI Similarity Search; the vector-similarity search library.
E5	Text-embedding model family used for dense representations.
API	Application programming interface.
BERT	Bidirectional Encoder Representations from Transformers; the encoder family used for BERTScore here.
IDF	Inverse document frequency; optional token weighting disabled in this study's BERTScore calculation.
Cross-encoder	Model that jointly processes a query and a candidate passage to assign a relevance score for reranking.
Chunk	A segment of a source document used as a retrieval unit.
Candidate pool	Retrieved chunks available before subsequent fusion or final selection.
RRF cutoff	Rank threshold used to retain candidates after reciprocal rank fusion.
Final context	Retrieved chunks actually supplied to the answer-generation model.
P0	Closed-book architecture without retrieved context.
P1	Dense-retrieval architecture.
P2	Hybrid dense and sparse retrieval with reciprocal rank fusion.
P3	Hybrid retrieval with cross-encoder reranking.